Parallel Successive Optimization for Unimodular Fractional Quadratic Programming

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***Abstract*—The Unimodular Fractional Quadratic Programming (UFQP) problem finds wide applications in signal processing, such as colocated Multiple-Input Multiple-Output (MIMO) radar and Multiple-Input Single-Output (MISO) secure communication. Conventional methods typically employ the Dinkelbach or coordinate descent (CD) framework. The Dinkelbach scheme results in a double-loop algorithm structure and the bottleneck lies in solving the subproblem. The CD approach advocates cyclic elementwise minimization of UFQP but becomes inefficient for large problem scales. Additionally, a closed-form minimization update is not yet available in the existing literature. To solve UFQP more efficiently, we propose an innovative algorithm based on the successive convex approximation (SCA) framework. We iteratively update all the variable components in parallel using closed-form expressions and develop a strictly feasible line search technique ensuring monotonic objective improvement. To avoid plateau region trapping, we introduce an escaping mechanism to find an update that yields a notable objective decrease based on majorization-minimization (MM).**

***Index Terms*—Unimodular optimization, fractional quadratic programming, parallel update, saddle point escape.**

I. Introduction

Unimodular Fractional Quadratic Programming (UFQP) arises from the practical need for Signal-to-Interference-plus-Noise Ratio (SINR) maximization and has a unified formulation as

$$\begin{aligned} \underset{\mathbf{x}\in\mathbb{C}^N}{\text{minimize}} \quad & f(\mathbf{x}) = \frac{\mathbf{x}^H\mathbf{A}\mathbf{x}}{\mathbf{x}^H\mathbf{B}\mathbf{x}} \\ \text{subject to} \quad & |x_n| = 1, n = 1, \ldots, N \end{aligned} \tag{1}$$

with $\mathbf{A} \succeq \mathbf{0}$ and $\mathbf{B} \succeq \mathbf{0}$. The variable $\mathbf{x}$ stands for the transmit waveform and the objective represents the reciprocal of SINR. The unimodular constraint allows transmitters to operate with maximum energy efficiency and prevents signal distortion in power amplifiers [1, 2].

The UFQP problem is motivated by various engineering scenarios in signal processing. In a colocated Multiple-Input Multiple-Output (MIMO) radar system, the SINR metric can be expressed as [3, 4, 5]

$$\frac{\mathbf{x}^H\left[\beta_0\left(\mathbf{I}\otimes\left[\mathbf{a}_t(\theta_0)\,\mathbf{a}_t^H(\theta_0)\right]\right)\right]\mathbf{x}}{\mathbf{x}^H\left[\sigma^2\mathbf{I}+\sum_{q=1}^{Q}\beta_q\left(\mathbf{I}\otimes\left[\mathbf{a}_t(\theta_q)\,\mathbf{a}_t^H(\theta_q)\right]\right)\right]\mathbf{x}} \tag{2}$$

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where $\beta_0, \beta_1, \ldots, \beta_Q$ are the reflection gains from θ_0 (the target angle), $\theta_1, \ldots, \theta_Q$ (angular directions of Q interfering scatterers), $\mathbf{a}_t(\theta_0), \mathbf{a}_t(\theta_1), \ldots, \mathbf{a}_t(\theta_Q)$ are steering vectors of uniform linear arrays, and σ^2 is the background noise power. The quadratic matrices in the numerator and denominator are both positive semidefinite (PSD), which matches the setting of problem (1). If all the scene parameters are known a priori, we can optimize the unimodular transmit waveform so that the SINR is maximized (the reciprocal is minimized) at the receiver end, which gives rise to a UFQP problem. In massive Multiple-Input Single-Output (MISO) downlink secure communication [6], the secrecy rate maximization problem is converted to a fractional formulation through the semidefinite relaxation (SDR) approach. In the case of single Bob (legitimate user) and single Eve (eavesdropper), the fractional objective can be written as

$$\frac{\mathbf{x}^H\left[\frac{1}{N}\mathbf{I}+P\mathbf{G}\mathbf{G}^H\right]\mathbf{x}}{\mathbf{x}^H\left[\frac{1}{N}\mathbf{I}+P\mathbf{h}\mathbf{h}^H\right]\mathbf{x}} \tag{3}$$

where P is the per-antenna transmit power, $\mathbf{G}$ denotes the channel from the transmitter to Eve, and $\mathbf{h}$ represents the channel to Bob. The received signal power of Bob should be increased while the signal captured by Eve should be suppressed to ensure security. Thus, the fractional objective (3) needs to be minimized. Additionally, the variable $\mathbf{x}$ is unimodular because the per-antenna transmit power is fixed. Thus, we obtain another UFQP problem.

A prevalent solution method for problem (1) is the Dinkelbach's algorithm which transforms the original fractional objective into a series of subtractive parametric subproblems. Yu et al. [3] and Li et al. [6] employed the Dinkelbach approach to UFQP and iteratively solve a unimodular quadratic program (UQP) using coordinate descent (CD) or alternating direction method of multipliers (ADMM). The solution algorithms are abbreviated as Dinkel-CD and Dinkel-ADMM, respectively. The Dinkelbach scheme creates a double-loop structure and the bottleneck lies in solving the UQP subproblem. The CD method only updates one variable at a time and is not desirable in a high-dimensional scenario. The ADMM framework may require a considerable number of iterations before termination and exhibit slow convergence near stationary points.

An alternative to the Dinkelbach scheme is the CD approach. Yu et al. [3] and Cui et al. [7] recommended cyclic elementwise minimization on the original UFQP and resort to the Dinkelbach-type algorithm for the minimizer of a scalar fractional function (CD-Dinkel). This approach becomes inefficient for long waveform codes and numerical pursuit of minimizers further increases the running time. Note that UFQP can also be solved via manifold optimization. The unimodular constraint defines a Riemannian manifold and unimodular-constrained problems can be addressed with the Riemannian conjugate gradient (RCG) algorithm. Hence, Hu et al. [4] applied RCG to solve UFQP. However, generic methods may not be able to exploit UFQP's characteristics.

In order to break the double-loop structure and solve UFQP with better efficiency, we develop a novel algorithm based on successive convex approximation (SCA) [8]. The technical novelty manifests in the following three aspects. Firstly, we implement a closed-form parallel update for all variable components, which proves to be more efficient than the cyclic update and iterative univariate minimization methods. Secondly, we incorporate a line search technique which maintains strict feasibility within the nonconvex constraint set and guarantees monotonic improvement of the objective function. Lastly, we introduce a dedicated majorization-minimization (MM)-based [9] mechanism in an attempt to escape saddle points when the algorithm is close to convergence.

II. Parallel Minimization Algorithm for UFQP

The proposed algorithm was inspired by the SCA framework [8]. For each variable block, SCA locally approximates the objective by a surrogate function and then minimizes the surrogate to produce a best-response solution. Best-response solutions of all blocks collectively form a descent direction with respect to the current iterate. A step size is then applied along this direction to update all the variable blocks. After an adequate number of iterations, the SCA framework produces a stationary point of the original problem.

In UFQP, we decompose the optimization variable elementwisely, so each block is a complex scalar. When the nth block is to be minimized, we separate x_n from the other elements in the UFQP objective:

$$\begin{aligned}\mathbf{x}^H\mathbf{A}\mathbf{x} =& A_{nn}\left|x_n\right|^2 + 2\mathrm{Re}\left[x_n^*\left(\sum_{m\neq n} A_{nm}x_m\right)\right]\\ &+\left(\sum_{m_1\neq n}\sum_{m_2\neq n} x_{m_1}^* A_{m_1m_2}x_{m_2}\right) \overset{(a)}{=} c_{A,n} + 2\mathrm{Re}\left[x_n^* c_{a,n}\right]\end{aligned} \tag{4}$$

$$\begin{aligned}\mathbf{x}^H\mathbf{B}\mathbf{x} =& B_{nn}\left|x_n\right|^2 + 2\mathrm{Re}\left[x_n^*\left(\sum_{m\neq n} B_{nm}x_m\right)\right]\\ &+\left(\sum_{m_1\neq n}\sum_{m_2\neq n} x_{m_1}^* B_{m_1m_2}x_{m_2}\right) \overset{(a)}{=} c_{B,n} + 2\mathrm{Re}\left[x_n^* c_{b,n}\right]\end{aligned} \tag{5}$$

$$\frac{\mathbf{x}^H\mathbf{A}\mathbf{x}}{\mathbf{x}^H\mathbf{B}\mathbf{x}} = \frac{c_{A,n} + 2\mathrm{Re}\left[x_n^* c_{a,n}\right]}{c_{B,n} + 2\mathrm{Re}\left[x_n^* c_{b,n}\right]} \tag{6}$$

where (a): due to the unimodular property $|x_n|^2 = 1$, we have $c_{A,n} = A_{nn} + 2\mathrm{Re}\left[x_n^*\left(\sum_{m\neq n} A_{nm}x_m\right)\right]$, $c_{B,n} = B_{nn} + 2\mathrm{Re}\left[x_n^*\left(\sum_{m\neq n} B_{nm}x_m\right)\right]$ (variable-independent constants), and $c_{a,n} = \sum_{m\neq n} A_{nm}x_m$, $c_{b,n} = \sum_{m\neq n} B_{nm}x_m$ (variable-dependent coefficients). As is shown in (6), if the other variable blocks are fixed, the UFQP objective reduces to a univariate fractional linear function. We minimize the fractional linear function, just like [3] and [7]:

$$\begin{aligned}&\underset{x_n\in\mathbb{C}^N}{\text{minimize}} && \frac{c_{A,n} + 2\mathrm{Re}\left[x_n^* c_{a,n}\right]}{c_{B,n} + 2\mathrm{Re}\left[x_n^* c_{b,n}\right]}\\ &\text{subject to} && |x_n| = 1.\end{aligned} \tag{7}$$

Note that no local approximation is needed, and this practice fits the SCA framework.

From the numerator of (6), we can obtain

$$\begin{aligned}& c_{A,n} + 2\mathrm{Re}\left[x_n^* c_{a,n}\right] = c_{A,n} + 2\mathrm{Re}\left[x_n^* c_{a,n}\right]\\ =& c_{A,n} + 2\left[\mathrm{Re}\left[x_n\right]\mathrm{Re}\left[c_{a,n}\right] + \mathrm{Im}\left[x_n\right]\mathrm{Im}\left[c_{a,n}\right]\right]\\ \overset{(a)}{=}& c_{A,n} + 2\left[\mathrm{Re}\left[c_{a,n}\right]\cos\theta_n + \mathrm{Im}\left[c_{a,n}\right]\sin\theta_n\right]\\ \overset{(b)}{=}& c_{A,n} + 2\left[\mathrm{Re}\left[c_{a,n}\right]\cdot\frac{1-t_n^2}{1+t_n^2} + \mathrm{Im}\left[c_{a,n}\right]\cdot\frac{2t_n}{1+t_n^2}\right]\end{aligned} \tag{8}$$

where (a) comes from the unimodular property and (b) is caused by half-angle substitution with $t_n \in \mathbb{R}$. We apply the same transformation to the denominator and the fractional linear function becomes

$$\begin{aligned}&\frac{c_{A,n} + 2\mathrm{Re}\left[x_n^* c_{a,n}\right]}{c_{B,n} + 2\mathrm{Re}\left[x_n^* c_{b,n}\right]} =\\ &\frac{c_{A,n}\left(1+t_n^2\right) + 2\left[\mathrm{Re}\left[c_{a,n}\right]\left(1-t_n^2\right) + \mathrm{Im}\left[c_{a,n}\right]\cdot 2t_n\right]}{c_{B,n}\left(1+t_n^2\right) + 2\left[\mathrm{Re}\left[c_{b,n}\right]\left(1-t_n^2\right) + \mathrm{Im}\left[c_{b,n}\right]\cdot 2t_n\right]}.\end{aligned} \tag{9}$$

We denote its minimum as λ_n. Since the numerator and denominator are both non-negative, for all $t_n \in \mathbb{R}$,

$$\begin{aligned}&\frac{c_{A,n}\left(1+t_n^2\right) + 2\left[\mathrm{Re}\left[c_{a,n}\right]\left(1-t_n^2\right) + \mathrm{Im}\left[c_{a,n}\right]\cdot 2t_n\right]}{c_{B,n}\left(1+t_n^2\right) + 2\left[\mathrm{Re}\left[c_{b,n}\right]\left(1-t_n^2\right) + \mathrm{Im}\left[c_{b,n}\right]\cdot 2t_n\right]} \geq \lambda_n\\ \Longleftrightarrow & c_{A,n}\left(1+t_n^2\right) + 2\left[\mathrm{Re}\left[c_{a,n}\right]\left(1-t_n^2\right) + \mathrm{Im}\left[c_{a,n}\right]\cdot 2t_n\right] \geq\\ & \lambda_n\left(c_{B,n}\left(1+t_n^2\right) + 2\left[\mathrm{Re}\left[c_{b,n}\right]\left(1-t_n^2\right) + \mathrm{Im}\left[c_{b,n}\right]\cdot 2t_n\right]\right)\\ \Longleftrightarrow & \left(p_{1,A,n} - \lambda_n p_{1,B,n}\right)t_n^2 + \left(p_{2,A,n} - \lambda_n p_{2,B,n}\right)\cdot t_n\\ & + \left(p_{3,A,n} - \lambda_n p_{3,B,n}\right) \geq 0\end{aligned} \tag{10}$$

where $p_{1,A,n} = c_{A,n} - 2\mathrm{Re}\left[c_{a,n}\right]$, $p_{1,B,n} = c_{B,n} - 2\mathrm{Re}\left[c_{b,n}\right]$, $p_{2,A,n} = 4\mathrm{Im}\left[c_{a,n}\right]$, $p_{2,B,n} = 4\mathrm{Im}\left[c_{b,n}\right]$, $p_{3,A,n} = c_{A,n} + 2\mathrm{Re}\left[c_{a,n}\right]$, and $p_{3,B,n} = c_{B,n} + 2\mathrm{Re}\left[c_{b,n}\right]$. Since there must exist a real-valued t_n satisfying the inequality, λ_n should be the maximum value for which the following system holds:

$$\begin{cases} p_{1,A,n} - \lambda p_{1,B,n} \geq 0\\ h\left(\lambda\right) \triangleq \left(p_{2,A,n} - \lambda p_{2,B,n}\right)^2\\ \qquad -4\left(p_{1,A,n} - \lambda p_{1,B,n}\right)\left(p_{3,A,n} - \lambda p_{3,B,n}\right) \leq 0.\end{cases} \tag{11}$$

In engineering practice, λ_n can be simply calculated as the larger root of $h\left(\lambda\right)$ that satisfies $p_{1,A,n} - \lambda p_{1,B,n} \geq 0$. With λ_n, we can obtain the best-response solution of x_n, denoted as $\hat{x}_n$:

$$\hat{x}_n = \exp\left(j\cdot 2\arctan\left[-\frac{p_{2,A,n} - \lambda_n p_{2,B,n}}{2\left(p_{1,A,n} - \lambda_n p_{1,B,n}\right)}\right]\right). \tag{12}$$

Closed-form minimization is more efficient than the iterative Dinkelbach-type method in [3] and [7].

In the classical SCA framework, a descent direction is construct by $\hat{\mathbf{x}} - \mathbf{x}$ where $\hat{\mathbf{x}}$ is a vector of all the best-response solutions $\{\hat{x}_n\}$ and the variable update is realized by $\mathbf{x}^+ = \mathbf{x} + \alpha\left(\hat{\mathbf{x}} - \mathbf{x}\right)$ with α being the step size. But in the UFQP context, the updated variable $\mathbf{x}^+$ may not be a feasible solution, so we modify the update rule as (the symbol definitions are consistent with those above)

$$\mathbf{x}^{+} = \mathbf{x} \odot (\hat{\mathbf{x}}/\mathbf{x})^{\alpha}. \tag{13}$$

Because unimodular variables preserve feasibility under multiplicative operations, $\mathbf{x}^{+}$ rigorously satisfies the constraint. The step size α is initialized to 1 and then decays recursively at a fixed rate until the condition $f(\mathbf{x}^{+}) \leq f(\mathbf{x})$ is met. This condition guarantees objective monotonicity with iteration.

III. Saddle Point Escape Technique

The UFQP has numerous saddle points where standard optimization algorithms may get stuck and terminate prematurely. As a result, saddle point escape techniques are necessary so as to navigate away from the plateau regions and move towards a better solution. When the convergence criterion is triggered, we attempt to further decrease the objective in a Dinkelbach manner:

$$\begin{aligned} \underset{\mathbf{y}\in\mathbb{C}^{N}}{\text{minimize}} \quad & g(\mathbf{y}) = \mathbf{y}^{H}\left[\mathbf{A} - \frac{\mathbf{x}^{H}\mathbf{A}\mathbf{x}}{\mathbf{x}^{H}\mathbf{B}\mathbf{x}}\mathbf{B}\right]\mathbf{y} \\ \text{subject to} \quad & |y_n| = 1, n = 1, \ldots, N. \end{aligned} \tag{14}$$

Since $g(\mathbf{x}) = 0$, an optimized output $\mathbf{y}$ can further decrease this value, i.e., $g(\mathbf{y}) \leq 0$, and thus $\frac{\mathbf{y}^{H}\mathbf{A}\mathbf{y}}{\mathbf{y}^{H}\mathbf{B}\mathbf{y}} \leq \frac{\mathbf{x}^{H}\mathbf{A}\mathbf{x}}{\mathbf{x}^{H}\mathbf{B}\mathbf{x}}$. Problem (14) is a UQP and this problem can be effectively solve with MM. We denote $\mathbf{A} - \frac{\mathbf{x}^{H}\mathbf{A}\mathbf{x}}{\mathbf{x}^{H}\mathbf{B}\mathbf{x}}\mathbf{B}$ as $\mathbf{C}$ and we know that $\mathbf{C} \preceq \kappa\mathbf{I}$ for any $\kappa \geq \lambda_{\max}(\mathbf{C})$. By Lemma 1 in [10], given any feasible point $\mathbf{y}_0$, the UQP objective is majorized by

$$\begin{aligned} &\mathbf{y}^{H}\mathbf{C}\mathbf{y} \\ \leq&\mathbf{y}^{H}(\kappa\mathbf{I})\mathbf{y} + 2\mathrm{Re}\left[\mathbf{y}^{H}(\mathbf{C} - \kappa\mathbf{I})\mathbf{y}_0\right] + \mathbf{y}_0^{H}(\kappa\mathbf{I} - \mathbf{C})\mathbf{y}_0 \\ =&2\mathrm{Re}\left[\mathbf{y}^{H}(\mathbf{C} - \kappa\mathbf{I})\mathbf{y}_0\right] + \left[\kappa N + \mathbf{y}_0^{H}(\kappa\mathbf{I} - \mathbf{C})\mathbf{y}_0\right]. \end{aligned} \tag{15}$$

A simple choice of κ is $\mathrm{Tr}(\mathbf{A})$ because $\mathbf{A} - \frac{\mathbf{x}^{H}\mathbf{A}\mathbf{x}}{\mathbf{x}^{H}\mathbf{B}\mathbf{x}}\mathbf{B} \preceq \mathbf{A} \preceq \lambda_{\max}(\mathbf{A})\mathbf{I} \preceq \mathrm{Tr}(\mathbf{A})\mathbf{I}$. Then we minimize the majorizing function to get the MM update:

$$\mathrm{MM}(\mathbf{y}_0) = -\exp\left(j \cdot \arg\left((\mathbf{C} - \kappa\mathbf{I})\mathbf{y}_0\right)\right). \tag{16}$$

Empirically, $\mathrm{MM}(\mathbf{y}_0)$ does not yield a significant objective decrease from $\mathbf{y}_0$, so we adopt the SQUAREM acceleration scheme [11] to magnify the progress:

$$\mathbf{y}_0^{+} = \mathrm{SQUAREM}(\mathbf{y}_0): \\ \begin{cases} \mathbf{y}_1 = \mathrm{MM}(\mathbf{y}_0), \mathbf{y}_2 = \mathrm{MM}(\mathbf{y}_1) \\ \mathbf{r} = \mathbf{y}_1 - \mathbf{y}_0, \mathbf{v} = \mathbf{y}_2 - \mathbf{y}_1 - \mathbf{r} \\ \beta = \|\mathbf{r}\|_2 / \|\mathbf{v}\|_2 \\ \text{while } g\left(\exp\left(j \cdot \arg\left(\mathbf{y}_0 + 2\beta\mathbf{r} + \beta^2\mathbf{v}\right)\right)\right) > g(\hat{\mathbf{y}}_0) \\ \quad \beta = (9\beta + 1)/10 \\ \mathbf{y}_0^{+} = \exp\left(j \cdot \arg\left(\mathbf{y}_0 + 2\beta\mathbf{r} + \beta^2\mathbf{v}\right)\right). \end{cases} \tag{17}$$

To this end, we summarize the proposed algorithm in Algorithm 1, named PSO. The PSO algorithm is built on the SCA framework, so stationarity convergence is guaranteed.

Computational Complexity Analysis: The analysis is carried out on a per-iteration basis. In the best-response derivation process, coefficients $c_{A,n}$, $c_{B,n}$, $c_{a,n}$, and $c_{b,n}$ for all n can be computed simultaneously with vectorized operations:

$$\begin{aligned} &\mathbf{a}_x = \mathbf{A}\mathbf{x}, \quad \mathbf{b}_x = \mathbf{B}\mathbf{x}, \quad D_A = \mathbf{x}^{H}\mathbf{a}_x, \quad D_B = \mathbf{x}^{H}\mathbf{b}_x, \\ &\mathbf{c}_a = \mathbf{a}_x - \mathrm{diag}(\mathbf{A}) \odot \mathbf{x}, \quad \mathbf{c}_b = \mathbf{b}_x - \mathrm{diag}(\mathbf{B}) \odot \mathbf{x}, \\ &\mathbf{c}_A = D_A \cdot \mathbf{1} - 2\mathrm{Re}\left[\mathbf{x}^{*} \odot \mathbf{c}_a\right], \mathbf{c}_B = D_B \cdot \mathbf{1} - 2\mathrm{Re}\left[\mathbf{x}^{*} \odot \mathbf{c}_b\right]. \end{aligned} \tag{18}$$

Thus, coefficients $p_{1,A,n}, p_{1,B,n}, p_{2,A,n}, p_{2,B,n}, p_{3,A,n}, p_{3,B,n}$, as well as $\hat{x}_n$ can be accordingly computed in parallel for all indices. The dominating operation is matrix-vector multiplication, of complexity $\mathcal{O}(N^2)$. The line search process requires repeated evaluation of the UFQP objective, which takes an $\mathcal{O}(N^2)$ complexity. When the saddle point escape mechanism is triggered, the MM update and UQP objective evaluation are the key operations. The primary computational load also stems from matrix-vector multiplication and has an $\mathcal{O}(N^2)$ complexity. Therefore, the per-iteration complexity of the proposed algorithm is $\mathcal{O}(N^2)$.

Algorithm 1 Parallel Successive Optimization (PSO) Algorithm for UFQP.

Require: Input: $\mathbf{A}$ and $\mathbf{B} \in \mathbb{C}^{N\times N}$. Iteration count $k = 0$. Escape trial count $t = 0$. Randomly initialize $\boldsymbol{\theta}_0$ from a uniform distribution over $[0, 2\pi)$ and $\mathbf{x}^{(0)} = \exp(j\boldsymbol{\theta}_0)$. Set $\gamma \in (0, 1)$ and total escape trials $T \in [1, 10]$. Specify termination thresholds ε_1 and ε_2.

repeat
  for $n = 1, \ldots, N$ **do**
    Compute $c_{A,n}$, $c_{B,n}$, $c_{a,n}$, and $c_{b,n}$ as described below (6)
    Compute $p_{1,A,n}$, $p_{1,B,n}$, $p_{2,A,n}$, $p_{2,B,n}$, $p_{3,A,n}$, and $p_{3,B,n}$ as described below (10)
    Obtain the best-response solution $\hat{x}_n^{(k)}$ as (12) with λ_n being the maximum satisfying (11)
  end for
  $\alpha = 1$
  while $f\left(\mathbf{x}^{(k)} \odot \left(\hat{\mathbf{x}}^{(k)}/\mathbf{x}^{(k)}\right)^{\alpha}\right) > f(\mathbf{x})$ **do**
    $\alpha = \gamma \cdot \alpha$
  end while
  $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} \odot \left(\hat{\mathbf{x}}^{(k)}/\mathbf{x}^{(k)}\right)^{\alpha}$
  if UFQP objective change $< \varepsilon_1$ and $t < T$ **then**
    Apply SQUAREM steps to iteratively solve UQP (14) ($\mathbf{x} = \mathbf{x}^{(k+1)}$ and $\mathbf{y} = \mathbf{x}^{(k+2)}$) until objective change $< \varepsilon_2$
  end if
  $k = k + 1$
until UFQP objective change $< \varepsilon_1$ and $t = T$

IV. Numerical Simulations

In this section, we present numerical results for the proposed PSO algorithm. All the experiments are conducted on a PC with a 2.90 GHz i7-10700 CPU and 16.0 GB RAM. We choose secure communication [6] as the simulation scenario. Let N and N_e be the number of antennas at the transmitter and the eavesdropper. We generate the parameters $\mathbf{A}$ and $\mathbf{B}$ according to equation (3): $\mathbf{A} = \frac{1}{N}\mathbf{I} + P\mathbf{G}\mathbf{G}^{H}$ and $\mathbf{B} = \frac{1}{N}\mathbf{I} + P\mathbf{h}\mathbf{h}^{H}$ where $P = 5$ and the entries of $\mathbf{G} \in \mathbb{C}^{N\times N_e}$ ($N = 200$ and $N_e = 30$) and $\mathbf{h} \in \mathbb{C}^{N}$ are drawn independently from the distribution $\mathcal{CN}(0, 1)$. The initial phases of the unimodular variable $\mathbf{x}$ are sampled independently and identically from a uniform distribution over $[0, 2\pi)$. The compared benchmark algorithms include Dinkel-CD [3], Dinkel-ADMM [6], CD-Dinkel [3, 7], and RCG [4].

When the problem dimension N is fixed at 200, we compare the convergence behavior of the aforementioned algorithms. We evaluate minimum objective gap (MOG) versus CPU time and the MOG metric is calculated as the difference between the second-to-last value of the objective trajectory and the minimum converged objective value of all the compared algorithms. Figure 1 shows that all the compared algorithms eventually reach a MOG level of less than 10^{-4} except CD-Dinkel, which suggests that CD-Dinkel suffers from inferior convergence precision and is prone to premature termination. Among the algorithms with high convergence precision (MOG less than 10^{-4}), the proposed PSO requires the least running time (0.14s), compared to the other benchmarks (Dinkel-ADMM: 0.19s, RCG: 0.33s, Dinkel-CD: 0.34s). Its CPU time is less than half that of the slowest algorithm.

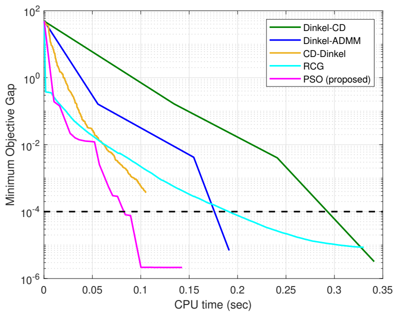

Fig. 1. Convergence behavior of the compared algorithms. Minimum objective gap (MOG) versus CPU time.

Then we vary the choice of N among $\{50, 100, 200, \ldots, 600\}$. The reported performance comes from the average of 100 randomized instances of UFQP. Figure 2 indicates that the converged MOG value of CD-Dinkel is consistently above 10^{-4} and is at least one order of magnitude larger than those of the other compared algorithms. The other algorithms generally achieve lower MOG values, except for the case of $N = 50$ where N is comparable to N_e. The converged MOG values of all the compared algorithms decrease with the problem size N. Dinkel-CD, Dinkel-ADMM, and PSO display better optimality performance and PSO is not as good as the other two when $N \geq 400$. But PSO is more efficient than Dinkel-CD and Dinkel-ADMM for all choices of N, as can be seen from the running time results in Figure 3. The CPU time of RCG is approximately equal to that of Dinkel-CD and Dinkel-ADMM. Among the algorithms with high convergence precision, PSO consumes the least computational time, which implies that the proposed algorithm achieves a favorable trade-off between optimality and efficiency.

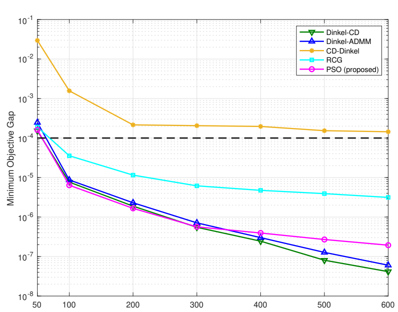

Fig. 2. The MOG metric versus problem dimension N.

Finally, we investigate into the effect of P on UFQP instances. The choice of P does not affect the computational time but influences the converged MOG. We vary P among $5 \times \{10^{-2}, 10^{-1}, \ldots, 10^{4}\}$ and the performance evaluation is presented in Figure 4. The converged MOG values exhibit a descending trend for all the compared algorithms. When P is close to zero, $\mathbf{A}$ and $\mathbf{B}$ are approximately scaled identity matrices. Three algorithms have a convergence precision greater than 10^{-4}, namely, CD-Dinkel, RCG, and PSO. PSO exceeds 10^{-4} by a narrow margin. The converged MOG's of Dinkel-CD and Dinkel-ADMM are around 10^{-5}. When the value of P is large, $\mathbf{A}$ and $\mathbf{B}$ are almost singular. The benchmark algorithms demonstrate MOG saturation (CD-Dinkel: 10^{-4}, RCG: 10^{-6}, Dinkel-CD: 10^{-7}, and Dinkel-ADMM: 10^{-7}) while the proposed PSO can reach a convergence precision below 10^{-9}.

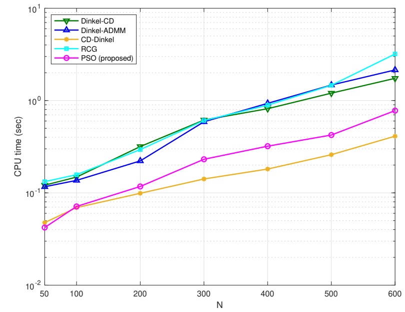

Fig. 3. Average running time versus problem dimension N.

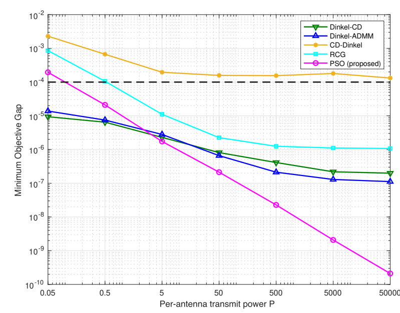

Fig. 4. The MOG metric versus per-antenna transmit power P.

V. Conclusion

In this paper, we have studied the solution algorithms of the UFQP problem. This problem constantly emerges in colocated MIMO radar and MISO secure communication. Traditionally, UFQP is solved with the Dinkelbach, CD, or RCG framework. The Dinkelbach scheme iteratively solves a UQP subproblem. The cyclic and non-analytical update of variable entries is not desirable for long waveform codes. RCG is a generic method to handle the unimodular constraint but may not fully capture the landscape of the objective function. To solve UFQP with better efficiency, we have put forward an SCA-based algorithm which enables closed-form parallel update for all variable components. For monotonic objective improvement, we have incorporated a strictly feasible line search operation. Moreover, we have designed an MM-based saddle point escape mechanism to further decrease the objective when the algorithm is close to convergence. Numerical simulations have shown that the proposed algorithm consumes the least computational time among the high-precision methods, indicating a favorable trade-off between optimality and efficiency.

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